A Survey and Empirical Evaluation of Missing Value Imputation Methods for Tabular Data

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ABSTRACT

Missing value imputation methods are commonly evaluated using reconstruction error metrics, such as root mean square error (RMSE). However, whether these metrics adequately reflect the practical utility of imputation methods in real-world applications remains insufficiently examined. In practice, imputation methods are often applied using default configurations, rather than tuning hyperparameters for each dataset.This study therefore examines the practical performance of imputation methods under default settings. We compare 13 imputation methods across 11 publicly available tabular datasets, resulting in 17,160 experiments covering three missing data mechanisms (MCAR, MAR, MNAR), two missing patterns, and four missing rates ranging from 10% to 50%. The results show that reconstruction accuracy does not necessarily align with downstream task performance. In terms of reconstruction accuracy, MissForest ranks among the best-performing methods (average rank = 2.06), yet its downstream classification and regression performance is only moderate. In contrast, Mean Imputation, despite its lower reconstruction accuracy, often outperforms more complex methods in downstream predictive tasks. This pattern is consistently observed across different missing mechanisms, patterns, and missing rates. Moreover, the five evaluated deep learning–based imputation methods do not consistently outperform Mean Imputation, whereas tree-based methods, such as MissForest and XGBoost-based imputation, demonstrate relatively stable overall performance across most settings.

Overall, if the objective is to accurately reconstruct missing values, tree-based methods remain a reliable choice. However, when the goal is to improve downstream predictive performance, imputation methods should not be evaluated solely based on reconstruction accuracy but rather on metrics aligned with the final application task.

Keywords

missing value imputation, tabular data, deep generative models, benchmark datasets, missing data mechanisms, survey

1. INTRODUCTION

In real-world data collection processes, missing data is a very common phenomenon. Missing values may arise for various reasons, such as equipment failures during data acquisition, respondents leaving survey questions unanswered, data loss during data integration, or errors in data recording. In many practical application domains, including medical data analysis, financial risk assessment, and social science research, missing data are often unavoidable. However, most statistical analysis techniques and machine learning algorithms typically assume that datasets are complete. As a result, the presence of missing values can significantly affect the reliability of statistical analysis and the performance of predictive models.

To address this issue, researchers have proposed a variety of missing value imputation methods that aim to estimate and fill in missing entries. By reconstructing incomplete datasets, these methods enable the data to be effectively used for subsequent analysis and model development.

1.1 What is Missing Value Imputation?

In data analysis, missing values refer to situations where the observed value of a variable is not recorded or cannot be obtained in a dataset[1]. Depending on how missing entries are distributed across the dataset, missing data can exhibit different missing data patterns. Common patterns include:(1) Univariate Pattern, where missing values occur only in a single variable while all other variables remain complete;(2) Monotone Pattern, where the dataset exhibits a staircase-like structure in which once a variable contains missing values, subsequent variables may also contain missing entries; and(3) General Pattern, where missing values are distributed irregularly across the dataset without a specific structure. This pattern is often the most common and also the most challenging to handle[1].In addition to missing patterns, statistical literature typically categorizes missing data according to the missing data mechanism. Three widely recognized mechanisms are:(1) Missing Completely at Random (MCAR), where the occurrence of missing values is independent of both observed and unobserved variables;(2) Missing at Random (MAR), where the missingness is related to other observed variables but not to the unobserved value of the variable itself; and(3) Missing Not at Random (MNAR), where the probability of missingness depends on the unobserved value itself[2].

To address the challenges introduced by incomplete data, researchers have developed missing value imputation techniques, whose primary goal is to estimate missing entries based on the observed data and generate a completed dataset. This allows the dataset to be effectively used for subsequent tasks such as statistical analysis, machine learning, and deep learning model training.

1.2 Why Missing Values Are a Problem?

The presence of missing values can have multiple impacts on data analysis and machine learning models. First, many statistical methods and machine learning algorithms assume that the input data are complete; therefore, when missing values exist in the dataset, these methods often cannot be directly applied. In addition, directly removing samples with missing values (e.g., listwise deletion) may substantially reduce the number of valid observations, thereby decreasing the statistical power of subsequent analyses. Second, inappropriate handling of missing values may distort the underlying data distribution. For example, simple imputation techniques may underestimate the variance of the data and disrupt the correlation structure among variables, which can affect the relationships learned by predictive models. Furthermore, when the missingness mechanism is related to the data itself, ignoring missing entries may lead to biased parameter estimates and unreliable analytical results. Such issues may ultimately degrade the reliability and predictive performance of data-driven models.

Therefore, developing methods that can reasonably estimate missing values while preserving the structural and statistical properties of the data has become an important research problem in data analysis, machine learning, and deep learning.

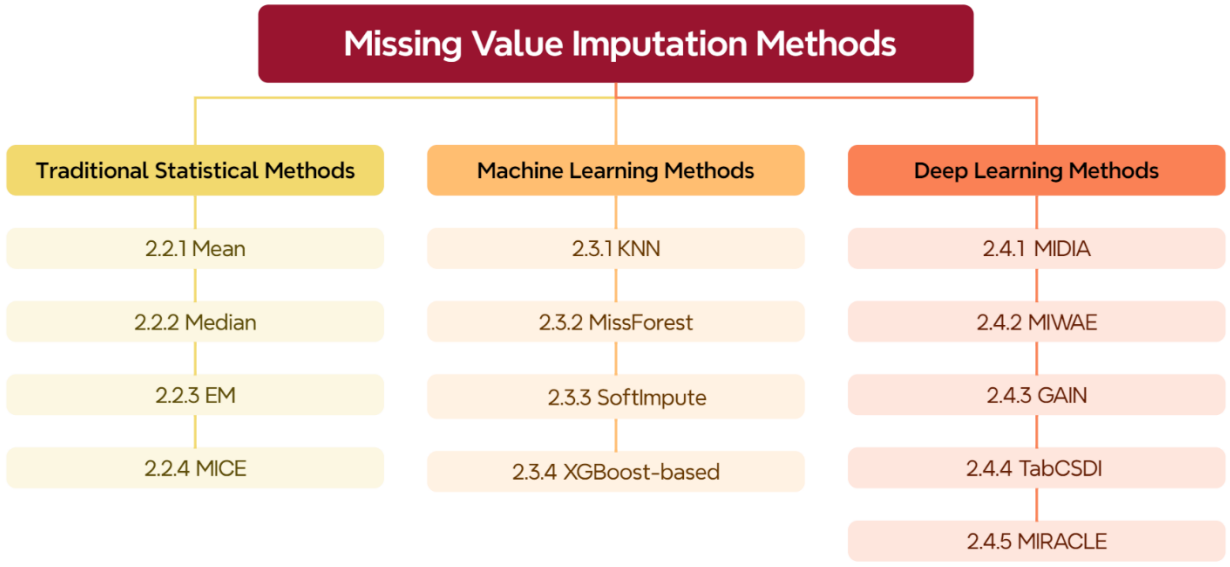


Figure 1: Taxonomy of missing value imputation methods reviewed in this survey.

1.3 Define the Goals of Survey

In recent years, with the rapid development of data science and artificial intelligence, a large number of methods have been proposed to address the problem of missing data. These approaches span a wide range of methodological paradigms, including traditional statistical techniques, machine learning methods, and deep learning models. However, these methods differ significantly in terms of their modeling assumptions, computational complexity, and applicable data scenarios. As a result, selecting an appropriate imputation method for a specific dataset remains a challenging task for researchers and practitioners. Therefore, this study aims to systematically review and compare existing missing value imputation methods, and to analyze their characteristics and limitations under different data conditions.

In this survey, missing value imputation methods are categorized into three major groups according to their modeling strategies: traditional statistical methods, machine learning methods, and deep learning methods. For traditional statistical methods, we review Mean Imputation, Median Imputation, Expectation–Maximization (EM), and Multiple Imputation by Chained Equations (MICE). For machine learning methods, we discuss KNN Imputation, MissForest, SoftImpute, and XGBoost-based Imputation. Finally, for deep learning methods, we review several representative approaches, including autoencoder-based imputation (MIDIA), variational generative model–based imputation (MIWAE), GAN-based imputation (GAIN), diffusion-based imputation (TabCSDI), and causal learning–based imputation (MIRACLE), in order to analyze how different deep learning frameworks model the missing value imputation problem. To compare the performance of different imputation methods, experiments are conducted on multiple publicly available datasets, and the performance of each method is evaluated using commonly adopted imputation evaluation metrics. The main contributions of this study are summarized as follows.

- We provide a systematic overview of the development of missing value imputation methods, and establish a taxonomy that categorizes approaches into statistical, machine learning, and deep learning frameworks.
- We analyze the modeling assumptions and methodological characteristics of different imputation techniques and compare their advantages and limitations under various data scenarios.
- We conduct empirical evaluations on multiple public datasets to compare representative imputation methods and provide practical insights for selecting appropriate imputation strategies in real-world applications.

Table 1: Literature Search Keywords Used in This Study

Category	Keywords
Topic Keywords	missing value imputation, missing data imputation
Method Keywords	KNN imputation, MissForest, matrix completion, softImpute, GAIN, MIWAE, TabCSDI

2. RELATED WORK

2.1 Literature Search and Selection Criteria

Missing values are an inevitable issue in data science and artificial intelligence applications. In real-world datasets, incomplete observations frequently occur due to various factors such as data collection errors, system limitations, or human omissions. Consequently, numerous studies have proposed different approaches to address this problem. These methods range from traditional statistical techniques to more sophisticated machine learning algorithms, and more recently to deep learning and generative AI–based solutions.

However, existing studies vary significantly in terms of data types, missingness mechanisms, and evaluation methodologies. Therefore, before conducting this survey, it is necessary to establish clear criteria for literature search and selection to ensure consistency and comparability across studies.In this study, we conducted a systematic literature search using Google Scholar. The search strategy consisted of two levels of keywords, as shown in Table 1.

The collected literature was further filtered based on several key criteria:(1)Relevance: The study must be directly related to missing value imputation.;(2)Representativeness:The study proposes a

representative method or is a highly cited seminal work.;(3)Completeness: The study clearly describes the datasets, missing data scenarios, experimental design, and evaluation metrics.

Through the above search and screening process, this study establishes a core set of literature on missing value imputation. The selected methods are further categorized into three main groups—traditional statistical methods, machine learning approaches, and deep learning methods—which serve as the foundation for the subsequent literature review and comparative analysis.

2.2 Traditional Statistical Imputation

Traditional statistical imputation methods represent the earliest approaches developed to handle missing data. The central idea of these methods is to estimate missing values using the statistical properties of the observed data or probabilistic models. In general, such approaches rely on measures of central tendency, distributional assumptions, or parameter estimation to perform imputation, and they have been widely used in early statistical analysis and data preprocessing.Compared with more recent machine learning and deep learning approaches, traditional statistical methods typically involve lower computational cost and provide clear statistical interpretability. However, they may face limitations when dealing with high-dimensional datasets or complex nonlinear relationships among variables.

This section introduces four representative statistical imputation methods: Mean Imputation (2.2.1) and Median Imputation (2.2.2), which rely on simple statistical estimates; Expectation–Maximization (EM) (2.2.3), which is based on maximum likelihood estimation; and Multiple Imputation by Chained Equations (MICE) (2.2.4), which operates under a multiple imputation framework. These methods represent three major modeling perspectives in statistical imputation: simple statistical estimation, parametric model–based estimation, and multiple-imputation inference.

2.2.1 Mean Imputation

Mean Imputation is one of the simplest and most commonly used methods for handling missing data. The basic idea of this approach is to replace missing values with the mean of the observed values for the corresponding variable. This method assumes that the variable exhibits a relatively stable central tendency, such that using the sample mean as an estimate can fill in missing entries without altering the overall mean of the dataset[1]. In practice, for each variable containing missing values, the mean of all observed entries is first computed, and this value is then assigned to all missing positions. Due to its extremely low computational cost and straightforward implementation, Mean Imputation is often used as a baseline method in studies on missing value imputation.

Despite its simplicity, Mean Imputation has several well-known limitations. In particular, replacing all missing entries with a single constant may substantially reduce the variance of the data and distort the correlation structure among variables. Because all imputed values are identical, the resulting distribution may become overly concentrated, which can negatively affect subsequent statistical analysis and model training. These limitations have motivated researchers to explore more advanced statistical methods that leverage the data distribution or relationships among variables to estimate missing values more appropriately.

2.2.2 Median Imputation

Median Imputation is conceptually similar to Mean Imputation in that it replaces missing values using a single summary statistic, but it uses the median of the observed values for the corresponding variable instead of the mean. Compared with the mean, the median is less sensitive to outliers, and therefore often provides a more robust estimate when the data distribution is skewed[1]. In practice, the median of all observed values for a variable is first computed, and this value is then used to replace the missing entries. Due to its simplicity and robustness to extreme values, Median Imputation has been widely used in early data analysis and preprocessing tasks.

However, Median Imputation shares several limitations with Mean Imputation. In particular, it may underestimate the variability of the data and fails to preserve the correlation structure among variables. Since the imputation relies solely on the statistical properties of a single variable, it does not utilize information from other features in the dataset. These limitations have motivated the development of more advanced statistical approaches that explicitly model the data distribution in order to estimate missing values more effectively.

2.2.3 Expectation-Maximization

The Expectation–Maximization (EM) algorithm was introduced by Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin [3]as a general method for performing maximum likelihood estimation in the presence of incomplete data or latent variables. The central idea of EM is to treat missing values as unobserved latent variables and iteratively update parameter estimates and missing data estimates, gradually approaching the maximum likelihood solution that would be obtained if the complete data were available.

The EM algorithm consists of two main steps: the Expectation step (E-step) and the Maximization step (M-step). In the E-step, the algorithm computes the conditional expectation of the missing data given the observed data and the current parameter estimates, thereby obtaining the expected value of the complete-data log-likelihood. In the M-step, the model parameters are updated by maximizing this expected log-likelihood. By iteratively alternating between the E-step and the M-step, the EM algorithm guarantees that the likelihood function monotonically increases at each iteration and eventually converges to a local maximum[3]. In the context of missing data problems, EM can be used to estimate the parameters of the data distribution and subsequently derive the expected values of missing entries. For instance, under a multivariate normal model, EM can estimate the mean vector and covariance matrix from the observed data and use these estimates to infer the expected values of the missing variables. Compared with simple statistical imputation methods such as mean or median imputation, EM can leverage the statistical relationships among variables, which in principle leads to more consistent and statistically meaningful estimates.

Despite these advantages, the EM algorithm also has several limitations. First, its convergence can be relatively slow in certain situations, particularly when dealing with high-dimensional datasets or high missing rates, where many iterations may be required before reaching a stable solution. Second, EM typically relies on distributional assumptions, such as the multivariate normal assumption; when the true data distribution deviates from these assumptions, the resulting estimates may be biased. Furthermore, EM generally produces a single imputed estimate, which does not directly capture the uncertainty associated with missing values. To address these issues, researchers later developed approaches such as Multiple Imputation, which aim to more fully account for the statistical uncertainty introduced by missing data.

2.2.4 Multiple Imputation by Chained Equations

Multiple Imputation by Chained Equations (MICE) was proposed by Stef van Buuren and Karin Groothuis-Oudshoorn[4]. It is a multiple imputation method based on fully conditional specification (FCS) for handling missing values in multivariate datasets. In multivariate imputation, common strategies include joint modeling (JM)

and fully conditional specification. Unlike joint modeling approaches, which require specifying a joint multivariate distribution for the data, MICE constructs variable-wise conditional models, providing a more flexible framework when it is difficult to reasonably assume a global distribution for the dataset.

The basic idea of MICE is to build a conditional imputation model for each variable with missing values and update these estimates iteratively. The algorithm typically begins by performing an initial imputation using simple methods. Then, each variable with missing entries is treated as a target variable in turn, and a regression model is constructed using the remaining variables as predictors to estimate the missing values. These conditional models are iteratively updated across variables, forming a sequence of chained equations. After several iterations, the algorithm gradually reaches stable imputation results. Unlike single imputation methods, MICE generates multiple completed datasets to reflect the uncertainty introduced by missing data. Researchers can perform standard analyses on each completed dataset and combine the results using Rubin's rules to obtain the final statistical estimates. Because each variable can adopt an imputation model appropriate to its data type (e.g., linear regression, logistic regression, or multinomial models), MICE is capable of handling both continuous and categorical variables. As a result, it has been widely applied in fields such as medical research, social sciences, and statistical data analysis.

Despite its flexibility, the performance of MICE depends heavily on the specification of the imputation models. Researchers must determine factors such as the model form, predictor variables, variable ordering, and the number of iterations, and inappropriate settings may affect imputation quality. In addition, the conditionally specified models do not necessarily correspond to a coherent joint distribution, which may introduce theoretical inconsistencies. Therefore, although MICE provides a highly flexible imputation framework, its effectiveness ultimately depends on appropriate model design and diagnostic procedures.

2.3 Machine Learning Based Imputation

Compared with traditional statistical approaches that rely on explicit distributional assumptions, machine learning–based methods reformulate missing value imputation as a prediction task, where missing entries are estimated by learning mappings among variables in a data-driven manner. By leveraging flexible model structures, these approaches are often able to capture nonlinear relationships and complex interactions among features.

This section reviews four representative machine learning–based imputation methods: the distance-based KNN Imputation (Section 2.3.1), the ensemble learning approach MissForest (Section 2.3.2), the matrix completion method SoftImpute (Section 2.3.3), and the gradient boosting–based XGBoost-based Imputation (Section 2.3.4). These methods represent four distinct modeling strategies for machine learning–based imputation: distance-based similarity estimation, iterative supervised learning, low-rank matrix approximation, and efficient ensemble prediction, respectively.

2.3.1 KNN Imputation

KNN Imputation was first proposed by Olga Troyanskaya et al. [5] who compared several missing value estimation methods on microarray datasets and recommended a KNN-based imputation strategy. The key motivation of this approach is that local estimation based on similar samples can be more robust than global statistical methods. Later, Gustavo Batista and Maria Carolina Monard [6] extended KNN Imputation to general data mining and machine learning scenarios.The core assumption of this method is that similar instances tend to have similar attribute values. For a target instance with missing values, the K-nearest neighbors algorithm identifies the K closest samples in the feature space and estimates the missing entries using the observed values of these neighbors. For continuous features, the imputed value is typically computed as the weighted average of the K neighbors. For categorical features, the most frequent category among the neighbors is used as the imputed value.

However, this approach faces several challenges in high-dimensional datasets due to the concentration of distances phenomenon. As the dimensionality increases, distances between samples tend to become similar, which reduces the discriminative power of distance-based similarity measures and makes neighbor selection less reliable. In addition, when the missing rate is high, the number of available features for computing distances decreases significantly, further reducing the reliability of neighbor identification. Experimental results reported by Troyanskaya et al. [5] show that KNN Imputation achieves lower RMSE than row-average and SVD-based methods when the missing rate is low to moderate (<10%). However, its advantage becomes much smaller when the missing rate exceeds 20%.

These limitations have motivated researchers to explore methods capable of capturing nonlinear relationships among features, rather than relying solely on distance-based similarity. This motivation led to the development of ensemble learning–based approaches such as MissForest, which are discussed in the next section.

2.3.2 MissForest

MissForest was proposed by Daniel J. Stekhoven and Peter Bühlmann[7]. The motivation of this method arises from the limitations of existing imputation approaches, which are often designed for a single data type or rely heavily on distributional assumptions. In contrast, MissForest aims to handle mixed-type data containing both continuous and categorical variables, while maintaining robust imputation performance in the presence of nonlinear relationships and feature interactions.

MissForest adopts a variable-wise iterative imputation strategy. The algorithm first initializes all missing entries with simple statistical estimates, such as the mean. The variables are then sorted according to their proportion of missing values, and imputation begins with the variable containing the smallest number of missing entries. For each target variable, the observed values are treated as the response, while the remaining variables serve as predictors. A random forest model is then trained to predict the missing entries of the target variable, where regression forests are used for continuous variables and classification forests for categorical variables. This process is repeated iteratively until convergence. Unlike KNN Imputation, which relies on local distance-based similarity, MissForest formulates the imputation task as a supervised learning problem that learns the conditional relationships between each variable and all other variables.

Despite its effectiveness, MissForest also has several limitations. One major drawback is its computational cost, since a random forest model must be repeatedly trained for each variable during the iterative imputation process. When the dataset contains a large number of variables or observations, the computational overhead can become substantial. In addition, Annie D. Shah et al. [8]pointed out that MissForest is designed to produce accurate point predictions for missing values rather than sampling from a posterior distribution, which may lead to biased parameter estimates in downstream statistical inference. Further simulation studies by Yue Hong and Henry Lynn [9]show that although MissForest often

achieves high imputation accuracy, it may produce biased regression coefficients and low confidence interval coverage under certain conditions, particularly when the data exhibit high skewness, nonlinear relationships, or outcome-dependent MAR mechanisms.

While the above methods perform imputation from the perspective of predicting missing values at the instance or variable level, SoftImpute approaches the problem from a different perspective by reconstructing the entire data matrix through low-rank approximation, which is discussed in the next section.

2.3.3 SoftImpute

SoftImpute was proposed by Rahul Mazumder, Trevor Hastie, and Robert Tibshirani [10]. Unlike prediction-based imputation methods that estimate missing values on a variable-by-variable basis, SoftImpute reformulates the missing data problem as a matrix completion task. Specifically, given a partially observed data matrix, the objective is to estimate a low-rank approximation capable of reconstructing the entire matrix.The core assumption of SoftImpute is that the observed data matrix is generated from an underlying low-dimensional latent structure, where high-dimensional observations can be approximated by a small number of latent factors. If the data matrix can be well approximated by a low-rank matrix, missing entries can be recovered by estimating this latent structure. In practice, the optimization objective of SoftImpute simultaneously minimizes the reconstruction error on the observed entries while encouraging the matrix to remain low-rank through nuclear norm regularization, controlled by a regularization parameter λ. The algorithm proceeds iteratively: in each iteration, missing entries are temporarily filled using the current estimate to form a completed matrix, after which a soft-thresholded singular value decomposition (SVD) is applied to update the low-rank approximation. The observed entries are preserved throughout the process until convergence.

The original experiments reported by Mazumder et al. [10]focused on large-scale sparse matrices, such as those arising in the Netflix recommendation system, demonstrating advantages in both computational efficiency and reconstruction accuracy. However, the low-rank assumption of SoftImpute implies that the relationships among variables are primarily linear, which may not hold when the underlying data relationships are highly nonlinear. In addition, the method is designed for numerical variables and cannot directly handle categorical data. When applied to mixed-type datasets, categorical variables typically need to be transformed using techniques such as one-hot encoding and converted back to their original form after imputation.

While the above three methods perform imputation from the perspectives of distance-based similarity, iterative supervised learning, and low-rank approximation, respectively, XGBoost-based Imputation combines an efficient gradient boosting framework with iterative imputation strategies to further improve the modeling capability for complex tabular data.

2.3.4 XGBoost-based

The theoretical foundation of XGBoost-based imputation can be traced back to Tianqi Chen and Carlos Guestrin [11], who introduced XGBoost as a highly scalable tree boosting system. Its key features include regularized boosting, sparsity-aware split finding, and efficient learning mechanisms designed for large-scale tabular datasets. It is worth noting that XGBoost was not originally designed for missing value imputation, but rather as a general-purpose gradient boosting framework for supervised learning.Building on this framework, Jing Deng and Thomas Lumley [12] proposed mixgb, which formally incorporates XGBoost into a multiple imputation framework. The method combines XGBoost with subsampling and predictive mean matching (PMM) to construct a scalable and robust imputation strategy. The motivation behind this approach is that traditional multiple imputation methods often rely heavily on the correct specification of imputation models and may suffer from limited computational efficiency or model expressiveness when applied to medium- or large-scale datasets with complex structures. In this framework, PMM generates imputed values by randomly sampling from observed values whose predicted values are close to the predicted missing value, thereby preserving the empirical distribution of the data while introducing reasonable uncertainty.

From a methodological perspective, XGBoost-based imputation treats each variable containing missing values as a supervised learning problem, using the remaining variables as predictors to train an XGBoost model that estimates the missing entries. This strategy is conceptually similar to the iterative variable-wise prediction framework used in MissForest. The key difference lies in the underlying ensemble learning technique: MissForest relies on bagging through random forests, whereas XGBoost employs boosting, which iteratively focuses on correcting residual errors and placing greater emphasis on difficult samples. As a result, boosting-based models may provide stronger modeling capability when complex feature interactions are present.Despite these advantages, XGBoost-based imputation also has several limitations. Compared with methods such as MissForest or SoftImpute, there is no single standardized algorithmic formulation for XGBoost-based imputation, and its performance may depend heavily on factors such as initialization strategies, the use of chained equations, the integration of predictive mean matching, and hyperparameter settings. Furthermore, although XGBoost natively supports missing values during tree construction, this mechanism essentially learns an optimal split direction for missing entries rather than explicitly reconstructing the missing values. If imputation is performed using only single-step predictions without incorporating mechanisms such as PMM, the uncertainty associated with imputed values may be underestimated.

While these machine learning approaches improve the accuracy and applicability of imputation through different modeling strategies, they still share a common limitation: the imputation process often relies on explicit feature engineering or predefined model structures, which may limit their ability to capture deeper latent structures in the data. With the rapid development of deep learning techniques, researchers have increasingly explored neural network–based imputation methods that aim to model the data distribution through end-to-end representation learning.

2.4 Deep Learning Based Imputation

With the rapid development of deep learning techniques, neural network–based models have increasingly been applied to missing value imputation problems. Compared with traditional machine learning methods that often rely on specific model structures or local relationships to estimate missing values, deep learning approaches can learn complex nonlinear relationships and latent representations through multi-layer neural networks. This capability provides stronger expressive power when dealing with high-dimensional data and complex data distributions.

In recent years, various deep learning–based imputation methods have been proposed, including representation learning approaches based on autoencoders, generative methods based on latent variable models, adversarial learning methods using GANs, and diffusion-based generative models.This section reviews four representative deep learning–based imputation methods: the denoising

autoencoder–based MIDIA (Section 2.4.1), the latent variable generative model MIWAE (Section 2.4.2), the adversarial learning approach GAIN (Section 2.4.3), and the diffusion-based method TabCSDI (Section 2.4.4). These methods represent four distinct modeling strategies for deep learning–based imputation: representation learning, latent variable generative modeling, adversarial generative learning, and diffusion-based generative modeling, respectively.

2.4.1 Missing Data Imputation Denoising Autoencoder

Traditional statistical and machine learning approaches for missing value imputation typically rely on explicit data assumptions or local similarity relationships to estimate missing entries. However, in real-world datasets, attributes often exhibit complex nonlinear dependencies, making it difficult for these approaches to effectively capture the underlying structure of the data. With the rapid development of deep learning, neural network–based models have increasingly been applied to representation learning tasks, providing new opportunities for addressing missing data problems.Among these models, the autoencoder (AE) is a widely used neural network architecture designed to learn compact representations of data. An autoencoder encodes input data into a lower-dimensional latent representation and then decodes this representation to reconstruct the original input. Through this process, the model learns intrinsic data characteristics and the relationships among attributes. Consequently, researchers have explored the use of autoencoders for missing value imputation.

In particular, the denoising autoencoder (dAE) introduces artificial corruption to the input data and trains the model to reconstruct the original clean data. This process encourages the model to learn robust representations that are resilient to noise and data sparsity[13]. Because the objective of a denoising autoencoder is to recover original information from partially corrupted inputs, it appears conceptually well suited for missing value imputation. However, directly applying dAEs to missing data problems still presents several limitations. First, the primary goal of dAEs is to learn stable and representative feature representations rather than explicitly estimating missing values. Second, their training objective typically focuses on reconstructing the entire input vector instead of specifically predicting the missing entries, which may lead to suboptimal performance in imputation tasks.

To address these limitations, Ma et al. proposed Missing Data Imputation Denoising Autoencoder (MIDIA)[14], a deep learning model specifically designed for missing value imputation. MIDIA estimates missing values by learning the implicit nonlinear relationships between observed and missing attributes. Furthermore, the model introduces two imputation strategies to accommodate different missing data patterns: MIDIA-Sequential and MIDIA-Batch. The MIDIA-Sequential approach trains a separate model for each attribute containing missing values and performs imputation sequentially, making it suitable for monotone missing patterns. In contrast, MIDIA-Batch employs a single model to impute all missing values simultaneously, which is more appropriate for general missing patterns. Experimental results demonstrate that MIDIA achieves higher imputation accuracy than several traditional statistical and machine learning methods on real-world datasets while effectively handling different data types and missing patterns.

2.4.2 Missing data Importance Weighted Autoencoder

Within deep generative modeling, deep latent variable models (DLVMs) have been widely recognized for their strong data generation capabilities. These models learn latent structures underlying the observed data and are therefore able to capture complex nonlinear relationships among variables. As a result, DLVMs have been considered promising approaches for missing value imputation. However, many existing DLVM-based methods assume that the training data are fully observed, which makes model training challenging when missing values are present in the dataset.

To address this issue, Antoine Mattei and Jes Frellsen proposed Missing data Importance Weighted Autoencoder(MIWAE)[15], a deep generative model designed to be trained directly on incomplete datasets. MIWAE is based on the Importance Weighted Autoencoder (IWAE) architecture and performs model learning by maximizing a lower bound of the observed data log-likelihood. By incorporating importance weighting, the method improves the approximation of the underlying data distribution and enables approximate maximum likelihood estimation under the missing at random (MAR) assumption. Compared with the original IWAE framework, MIWAE can handle datasets with missing values without introducing additional computational overhead, making it computationally efficient in practical applications.

Furthermore, MIWAE employs Monte Carlo sampling techniques to enable both single imputation and multiple imputation using the trained model. Through this mechanism, missing values are inferred from the learned latent data distribution rather than relying solely on local similarity or simple statistical assumptions. This allows MIWAE to preserve the underlying generative structure of the data during the imputation process.Experimental results demonstrate that MIWAE achieves highly accurate imputation performance on several continuous datasets and remains competitive with multiple state-of-the-art methods. In addition, the authors showed that convolutional DLVMs can be successfully trained on incomplete versions of the MNIST dataset, further illustrating the feasibility of the approach for high-dimensional data scenarios.

2.4.3 Generative Adversarial Imputation Nets

In the literature on missing value imputation, existing approaches are often categorized into discriminative methods and generative methods. Discriminative methods, such as MICE, MissForest, and matrix completion–based approaches, typically estimate missing values by learning mappings between input variables and target variables. While these approaches have demonstrated practical effectiveness, they may struggle to capture complex distributional structures in data, particularly in high-dimensional settings or when strong nonlinear relationships exist among variables.

In contrast, generative methods aim to perform imputation by modeling the joint distribution of the data. Classical approaches such as Expectation Maximization (EM) fall into this category. Although these methods are theoretically capable of better representing the underlying data distribution, they also present several limitations. Some approaches require strong assumptions about the data distribution, which may limit their applicability when datasets contain mixed variable types (e.g., categorical and continuous variables). Furthermore, several methods based on denoising autoencoders (DAEs) have shown promising empirical performance but often require complete datasets during training. In many real-world applications, however, missing values are inherently part of the data generation process, and obtaining a fully observed dataset is often unrealistic. Even when certain models can be trained on incomplete datasets, they may rely primarily on observed values to learn representations while failing to effectively leverage missingness information. In addition, some GAN-based approaches developed for image completion require fully observed data for training the

discriminator, which makes them difficult to directly apply to general missing data imputation tasks.

To address these challenges, Jinsung Yoon et al. proposed Generative Adversarial Imputation Nets (GAIN)[16], a missing data imputation framework based on Generative Adversarial Networks (GANs). The model estimates missing values through adversarial learning between a generator and a discriminator. The generator attempts to infer missing entries conditioned on the observed components of the data vector, while the discriminator tries to determine whether each element in the completed data vector corresponds to an originally observed value or an imputed value produced by the generator. Through this adversarial training process, the model gradually learns the underlying distribution of the data.In addition, GAIN introduces a hint mechanism into the standard GAN framework[16]. This mechanism provides the discriminator with partial information about the missingness pattern, allowing it to focus on the dimensions that are most relevant to evaluating imputation quality. Such a design encourages the generator to produce imputations that better match the true data distribution. Experimental results demonstrate that GAIN outperforms several existing imputation methods across multiple benchmark datasets[16], highlighting the potential advantages of adversarial learning for handling missing data problems.

2.4.4 Conditional Score-based Diffusion Models for Tabular data

With the continued development of missing value imputation research, recent studies have increasingly explored the use of deep generative models for handling incomplete data, in addition to traditional iterative approaches. While iterative methods estimate conditional distributions and repeatedly update imputed values, generative models instead learn the joint distribution of the data and generate the missing components conditioned on the observed entries. Representative studies in this category include MIDA, HIVAE, MIWAE, and GAIN. Because deep generative models are capable of capturing complex nonlinear relationships in data, they have attracted growing attention for missing value imputation tasks.More recently, diffusion models have achieved remarkable success in generative modeling and have been widely applied to tasks such as image generation, time-series modeling, molecular generation, and natural language processing. Compared with traditional generative models such as GANs or VAEs, diffusion models learn the data distribution through a process of gradually adding and removing noise. This formulation often leads to more stable training and higher-quality generation. Despite their success in many generative tasks, however, research on applying diffusion models to missing value imputation for tabular data remains relatively limited.

To address this gap, Nontawat Charoenphakdee et al. proposed Conditional Score-based Diffusion Models for Tabular Data (TabCSDI)[17], a diffusion-based approach designed for missing value imputation. The method is inspired by the Conditional Score-based Diffusion Model for Imputation (CSDI)[18], which was originally developed for missing data imputation in time-series datasets. However, since time-series datasets typically consist of numerical features only, the original CSDI model does not support categorical variables. In contrast, tabular datasets often contain both numerical and categorical attributes. To address this limitation, TabCSDI modifies the model architecture to enable the joint modeling of both data types.To effectively handle categorical variables, the authors further investigate and compare three different data representation strategies: one-hot encoding, analog bits encoding, and feature tokenization. Experimental results on multiple benchmark datasets show that TabCSDI can successfully learn the underlying data distribution and perform missing value imputation[17]. Its performance is competitive with existing approaches from both iterative and generative imputation frameworks. In addition, the results indicate that categorical embedding methods have a significant impact on model performance, highlighting the importance of feature representation in tabular data imputation tasks.

2.4.5 Missing data Imputation Refinement And Causal Learning

Most missing value imputation methods primarily focus on learning the data distribution and improving imputation accuracy. However, these approaches often overlook the causal structure underlying the data. When missing data are present, relying solely on observed data for imputation may introduce spurious correlations, leading to a distribution shift between the observed data distribution and the true complete data distribution. Such discrepancies can negatively affect the reliability of subsequent analyses and predictive models.To address this issue, Jie Zhang and Mihaela van der Schaar proposed Missing Data Imputation Refinement and Causal Learning (MIRACLE)[19], a causal learning–based framework for missing value imputation. The method is motivated by the perspective of causal inference, which suggests that the imputation process should preserve the causal dependencies present in the underlying data-generating mechanism. Traditional imputation methods typically use all observed variables to predict missing entries. In contrast, MIRACLE estimates the conditional expectation of a missing variable using only its causal parents in the causal graph[19], thereby reducing the risk of incorporating spurious correlations introduced by the missingness mechanism.

MIRACLE is designed as a general imputation framework that can be applied on top of existing imputation methods. Specifically, the approach first employs a baseline imputation method to generate initial imputations. It then learns a missingness graph (m-graph) to model the data generation process associated with missingness. Based on this structure, the method introduces causal regularization into the imputation function, encouraging the imputed values to remain consistent with the causal relationships in the data. Through this iterative refinement process, MIRACLE gradually improves the imputation results and enhances overall imputation quality.

The proposed framework was evaluated on both synthetic datasets and several benchmark datasets from the UCI Machine Learning Repository[19]. In the experiments, MIRACLE was applied as a refinement framework on top of six commonly used imputation methods. The results demonstrate that MIRACLE consistently improves the performance of baseline imputation methods in most scenarios, while maintaining stable performance across different missing data mechanisms, including missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR).

3. METHODOLOGY

3.1 Experimental Pipeline

As illustrated in **Figure 2**, the experimental procedure of this study consists of several steps. First, the original datasets are loaded and rows containing naturally occurring missing values are removed to obtain a complete subset that serves as the ground truth. The dataset is then split into training and testing sets using an 80/20 split, where stratified sampling is applied for classification tasks to preserve the class distribution.Next, preprocessing is performed on the datasets. Numerical features are scaled to the range [0, 1] using Min–Max scaling, with statistics computed from the training set only. Categorical variables are encoded using label encoding. Both transfor-

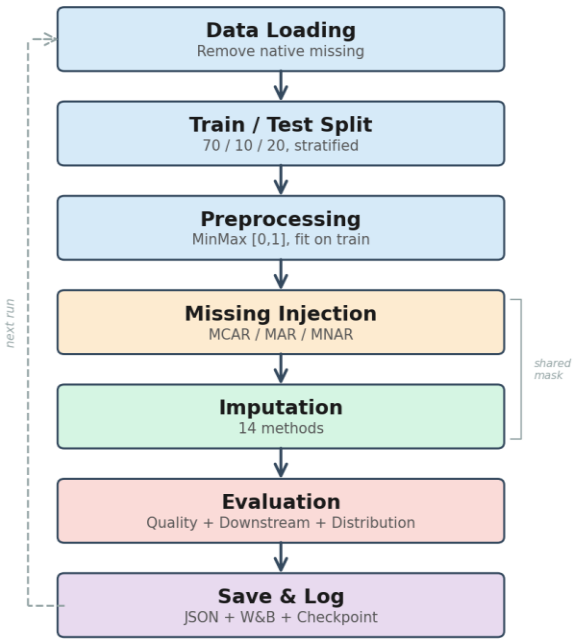


Figure 2: Experimental pipeline for evaluating missing value imputation methods.

mations are fitted exclusively on the training set to avoid data leakage.

After preprocessing, artificial missing values are injected into both the training and test sets according to the specified missing mechanisms, missing patterns, and missing rates, producing corresponding missing datasets and missing masks. Each imputation method is first fitted on the training set and then used to impute missing values in both the training and test sets. Finally, the imputation results are evaluated using three categories of evaluation metrics, as described in **Section 4.2**.

3.2 Missing Data Simulation

Missing data simulation in this study involves three dimensions: missing mechanisms, missing patterns, and missing rates.

Missing Mechanisms. We consider the three standard missing data mechanisms commonly discussed in the literature: (1) MCAR (Missing Completely At Random), where each value is independently masked with a fixed probability $\boldsymbol{p}$, and the occurrence of missingness is unrelated to the values of any variables, representing an idealized baseline scenario; (2) MAR (Missing At Random), where the probability of missingness in a target variable depends on other observed variables. In this study, this dependency is modeled using a logistic function: an observed variable X_j is first standardized and then used to generate the missing probability for a target variable X_k , where the intercept is calibrated using binary search so that the overall missing rate approximates the desired level; and (3) MNAR (Missing Not At Random), where the probability of missingness depends on the value of the variable itself. The simulation procedure is similar to MAR, except that the logistic function uses the value of the target variable as input rather than another variable, making MNAR the most challenging scenario among the three mechanisms. Including all three mechanisms allows us to examine the robustness of different imputation methods under varying assumptions about the missing data process. In particular, methods such as MIRACLE, which are designed for MNAR settings, can be evaluated to determine whether their theoretical advantages are reflected in empirical performance.

Missing Patterns. Two missing patterns are considered: (1) Univariate Pattern, where missing values are injected into a single variable with the largest variance in the dataset. This pattern represents a relatively simple scenario and is used to evaluate the basic imputation capability of each method; and (2) General Pattern, which more closely reflects real-world datasets. Under this pattern, missing values may occur across all variables in the dataset, and missing entries are injected independently according to the specified missing mechanism and missing rate. To ensure that each sample remains usable for downstream tasks, at least one observed feature is preserved in every row.

Missing Rates. To evaluate the robustness of imputation methods under different levels of data incompleteness, four missing rates are considered: 10%, 20%, 30%, and 50%, covering a range from low missingness to extremely high missingness. These settings allow us to analyze how the performance of different methods degrades as the missing rate increases.

3.3 Fairness Protocol

To ensure that the 13 imputation methods are compared under consistent conditions, the following fairness protocol is adopted.

Data and configuration consistency. Under the same(datatset,mechanism,pattern,rate,seed) configuration, all methods share the exact same missing mask, ensuring that each method faces identical missing positions. In addition, all methods use the same train/test split and identical scaling parameters. Numerical features are scaled using Min–Max scaling, where the scaling parameters are fitted only on the training set to avoid introducing data leakage or unfair variations caused by preprocessing differences.

Repeated experiments and statistical reporting. For each configuration, experiments are repeated with five different random seeds, and the results are reported as mean $\pm$ standard deviation to ensure the statistical robustness of the conclusions.

4. EXPERIMENT AND EVALUATION

4.1 Datasets Description

To evaluate the performance of different missing value imputation methods, this study conducts experiments on several publicly available tabular datasets. These datasets are primarily obtained from widely used machine learning data platforms such as the UCI Machine Learning Repository and Kaggle. They have been extensively used in previous research on machine learning and missing value imputation.

The selected datasets were chosen for several reasons. First, they provide well-structured tabular data that are suitable for evaluating imputation algorithms. Second, they vary in dataset size, feature dimensionality, and data characteristics, allowing a more comprehensive evaluation of imputation methods across different scenarios.Since the feature types of a dataset may significantly influence the design and effectiveness of imputation methods, the datasets in this study are categorized into two groups:(1) Pure numerical datasets, and(2) Mixed Numerical and Categorical Datasets.This classification is commonly adopted in tabular data research because different feature types often require different imputation strategies. For example, purely numerical datasets can often be handled using distance-based or distribution-based estimation methods, whereas mixed-type datasets require handling both categorical and numerical variables simultaneously. Some deep learning–based methods

Table 2: Dataset information

Datasets	#Categorical	#Numerical	#Samples
Wine	0	11	4,898
Concrete	0	8	1,030
Spam	0	57	4,601
Yeast	0	8	1,484
Breast	0	9	699
Letter	0	16	20,000
Adult	9	6	48,842
Bank Marketing	9	7	45,211
Titanic	5	5	891
Mamm. Masses	3	2	961
Autism	15	6	704

also require additional mechanisms to properly represent categorical features.

In total, 11 benchmark datasets are used in our experiments, including six purely numerical datasets and five mixed-type datasets. These datasets vary in both the number of samples and the number of attributes, ranging from hundreds to tens of thousands of instances. Such diversity enables us to examine the generalization ability of different imputation methods across datasets with varying scales and feature structures.The basic statistical information of each dataset, including the number of categorical attributes, numerical attributes, and samples, is summarized in the **Table 2**.

4.1.1 Pure Numerical Datasets

To evaluate the performance of different missing value imputation methods on purely numerical data, this study selects six publicly available datasets that contain only numerical attributes: Wine Quality, Concrete Compressive Strength, Spam, Yeast, Breast Cancer Wisconsin, and Letter Recognition. These datasets are obtained from publicly accessible repositories and are widely used in machine learning and missing value imputation research.Since these datasets contain only numerical variables, they allow us to directly evaluate the estimation capability of different imputation methods without being affected by categorical encoding or discrete feature processing. In addition, the selected datasets vary in both sample size and feature dimensionality, enabling a comprehensive evaluation of imputation methods under different data scales and feature structures.

1) Wine:This study uses the Wine Quality dataset from the UCI Machine Learning Repository. The dataset contains physicochemical properties related to wine quality, including attributes such as acidity, residual sugar, chlorides, density, pH, and alcohol. It consists of 4,898 samples and 12 attributes, making it a medium-sized tabular dataset. Due to its clear structure and frequent use in regression and missing data studies, it serves as a representative numerical benchmark dataset.

2) Concrete:The Concrete Compressive Strength dataset, also from the UCI Machine Learning Repository, records the relationship between concrete material composition and compressive strength. Its attributes include cement, slag, ash, water, superplasticizer, coarse aggregate, fine aggregate, and age, all of which are numerical variables. The dataset contains 1,030 samples and 8 attributes and is commonly used in regression and imputation studies due to its relatively simple structure.

3) Spam:The Spam dataset from the UCI repository is included as a high-dimensional numerical dataset. Originally developed for email spam detection, the features mainly consist of word frequency statistics, character frequency measures, and other numerical descriptors. The dataset contains 4,601 samples and 57 attributes, which makes it relatively high-dimensional compared with other datasets in this study. It is therefore useful for evaluating the robustness of imputation methods on high-dimensional numerical data.

4) Yeast:The Yeast dataset from the UCI repository was originally used in bioinformatics applications, where the attributes describe numerical characteristics related to protein localization and biological properties. The dataset contains 1,484 samples and 8 attributes, representing a medium-sized dataset with relatively simple feature structure. Its lower dimensionality makes it suitable for evaluating imputation performance on simpler numerical datasets.

5) Breast:This study also includes the Breast Cancer Wisconsin dataset from the UCI repository, which is a widely used benchmark in machine learning research. The attributes are numerical descriptors extracted from cell nucleus images, such as radius, texture, perimeter, and area. The dataset contains 699 samples and 9 attributes. Although the dataset is relatively small, its extensive use in previous studies makes it valuable for comparison with existing research.

6) Letter:The Letter Recognition dataset from the UCI repository is also included. This dataset originates from an image recognition task in which each letter is represented by a set of numerical features extracted from image characteristics. The dataset contains 20,000 samples and 16 attributes, making it one of the largest numerical datasets used in this study. Its relatively large sample size enables the evaluation of imputation methods on larger-scale numerical datasets.

Overall, these six numerical datasets differ in terms of sample size, number of attributes, and feature dimensionality, covering small, medium, and relatively large tabular datasets as well as both low- and high-dimensional numerical data. Such diversity allows a more comprehensive evaluation of the applicability and performance of different missing value imputation methods on purely numerical datasets.

4.1.2 Mixed Numerical and Categorical Datasets

In addition to purely numerical datasets, this study also includes several mixed-type datasets containing both numerical and categorical attributes. Compared with purely numerical data, mixed-type datasets generally present greater challenges for missing value imputation because different types of variables often require different imputation strategies. For instance, numerical variables are typically estimated using statistical or distance-based approaches, whereas categorical variables may require probabilistic models, classification-based techniques, or specialized generative models.

In recent years, several deep learning approaches—such as diffusion-based models for tabular data—have introduced additional mechanisms to properly handle categorical variables. Therefore, evaluating imputation methods on mixed-type datasets allows us to better reflect real-world data scenarios and examine the generalization ability of different imputation approaches.In this study, we select five mixed-type datasets: Adult, Bank Marketing, Titanic, Mammographic Masses, and Autism. These datasets vary in both dataset size and feature composition, ranging from several hundred to tens of thousands of samples and containing different proportions of numerical and categorical attributes. Such diversity allows a

more comprehensive evaluation of the performance of imputation methods on mixed-type tabular datasets.

1) Adult:This study uses the Adult dataset, also known as the Census Income dataset, from the UCI Machine Learning Repository. The dataset originates from U.S. census data and aims to predict whether an individual's income exceeds a certain threshold based on demographic and socioeconomic attributes. The features include variables such as age, education, occupation, marital status, and workclass, which consist of both numerical and categorical variables. The dataset contains 48,842 samples, including 9 categorical features and 6 numerical features. Due to its large size and widespread use in tabular machine learning research, it is commonly used as a benchmark dataset for evaluating models and data preprocessing techniques.

2) Bank Marketing:The Bank Marketing dataset, also from the UCI Machine Learning Repository, is related to direct marketing campaigns conducted by a Portuguese banking institution. The task is to predict whether a client will subscribe to a term deposit. The dataset contains demographic and financial attributes such as age, job, marital status, education, and account balance, including both numerical and categorical variables. It consists of 45,211 samples, with 9 categorical features and 7 numerical features. Due to its relatively large size and diverse feature types, the dataset is frequently used to evaluate tabular machine learning models in real-world business scenarios.

3) Titanic:The Titanic dataset originates from a well-known machine learning competition hosted on Kaggle. It records passenger information from the Titanic disaster and aims to predict whether a passenger survived the accident. The dataset includes attributes such as age, fare, sex, passenger class (Pclass), and embarked port, which consist of both numerical and categorical variables. The dataset contains 891 samples, including 5 categorical features and 5 numerical features. Due to its clear structure and moderate dataset size, it is commonly used for educational purposes and experimental studies in machine learning.

4) Mammo. Masses:The Mammographic Masses dataset from the UCI Machine Learning Repository is used for breast mass diagnosis analysis. The dataset contains features describing patient age and image-based diagnostic attributes, such as mass shape, mass margin, and density. These attributes include both numerical and categorical variables. The dataset contains 961 samples, including 3 categorical features and 2 numerical features. Although relatively small in size, the dataset has a clear structure and is frequently used in medical data analysis research.

5) Autism:The Autism dataset, also obtained from the UCI Machine Learning Repository, is used in autism screening research. The dataset includes multiple questionnaire-based assessment items and demographic information such as age, gender, ethnicity, and family history. It contains 704 samples, including 15 categorical features and 6 numerical features. Since the dataset contains a relatively high proportion of categorical variables, it is useful for evaluating the performance of imputation methods when handling categorical attributes.

Overall, these five mixed-type datasets cover a range of dataset sizes and feature compositions, including large-scale demographic datasets, medium-sized commercial datasets, and smaller medical and survey-based datasets. By conducting experiments on these datasets, we can more comprehensively evaluate the applicability and performance differences of various missing value imputation methods when applied to mixed-type tabular data.

4.2 Evaluation Metrics

To comprehensively and fairly evaluate the performance of different missing value imputation methods, this study adopts an evaluation framework consisting of three aspects: (1) Imputation Quality, which directly measures the error between imputed values and the corresponding ground truth; (2) Downstream Task Performance, which evaluates how well the imputed data support subsequent predictive tasks; and (3) Distribution Preservation, which examines whether the statistical distribution of the original data is preserved after imputation.

It is important to note that all imputation quality and distribution preservation metrics are computed only on the artificially masked entries, rather than on the entire dataset. This ensures that the evaluation reflects the model's ability to estimate missing values rather than its ability to reproduce already observed data.

4.2.1 Imputation Quality Metrics

For numerical variables, this study adopts two commonly used error metrics: Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

Root Mean Squared Error (RMSE). RMSE assigns a larger penalty to large errors and is one of the most widely used metrics in missing value imputation research, which facilitates comparison with prior studies.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \quad (1)$$

Mean Absolute Error (MAE). MAE penalizes all errors equally and is generally more robust when the data contain outliers. Using both metrics provides complementary perspectives and improves the robustness of the evaluation.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \quad (2)$$

For categorical variables, Accuracy and Macro-F1 are used. Accuracy measures the proportion of correctly imputed categorical values and provides an intuitive evaluation of prediction correctness; however, it may be misleading when the class distribution is imbalanced. Macro-F1 evaluates each class with equal weight and therefore provides a more balanced assessment of imputation performance, particularly for minority classes.

4.2.2 Downstream Task Performance

To evaluate how well the imputed data support downstream predictive tasks, this study employs a Random Forest model with fixed hyperparameters ($n_estimators = 100$) as the unified downstream learner, thereby avoiding additional comparison bias introduced by model selection. Downstream performance is evaluated according to the task type: (1) Regression tasks (Wine Quality and Concrete) are evaluated using R^2 and RMSE; (2) Binary classification tasks (Spambase, Breast Cancer, Adult, Bank Marketing, Titanic, Mammographic Masses, and Autism) are evaluated using Accuracy and AUC–ROC; and (3) Multi-class classification tasks (Yeast and Letter Recognition) are evaluated using Accuracy and Macro-F1.

4.2.3 Distribution Preservation

To evaluate whether the statistical properties of the original data are preserved after imputation, distribution-based metrics are employed for both numerical and categorical variables. For numerical variables, Wasserstein Distance is used to measure the transportation cost between the original distribution and the imputed distribution. Compared with KL Divergence, Wasserstein Distance is

symmetric and remains well-defined even when the supports of the two distributions do not fully overlap, resulting in more stable computation.

For categorical variables, Jensen–Shannon Divergence (JSD) is adopted. JSD is a symmetrized version of KL Divergence with a bounded range of $[0,1]$, where 0 indicates that the two distributions are identical. For both metrics, the distance is computed independently for each feature, and the arithmetic mean across all features is reported as the overall distribution preservation score.

4.3 Experiment Implementation

This section describes the implementation details of the experiments, including the experimental scale, software libraries used for implementing the imputation methods, hyperparameter configurations, and the hardware environment used in this study.

4.3.1 Experimental Matrix

In total, 17,160 experimental runs were conducted, corresponding to the full factorial combination of 13 imputation methods, 11 datasets, 3 missing mechanisms, 2 missing patterns, 4 missing rates, and 5 random seeds.

4.3.2 Hyperparameter Settings

To ensure a fair comparison across methods, all approaches are evaluated using fixed hyperparameters without dataset-specific tuning. The hyperparameter settings for statistical and machine learning methods are listed in **Table 3,** while those for deep learning methods are provided in **Table 4**.

4.3.3 Handling Categorical Variables

Since different imputation methods vary in their support for categorical variables, this study adopts method-specific strategies for handling categorical features. Mean, Median, EM, and SoftImpute are designed for numerical variables only; therefore, categorical columns are imputed using mode imputation as a fallback.

MissForest and MICE natively support mixed-type data. For KNN, XGBoost, MIDA, MIWAE, GAIN, and MIRACLE, categorical variables are first transformed using one-hot encoding, and the imputed results are converted back to the original categories using argmax. TabCSDI natively supports mixed-type data; however, in the main experiments, categorical variables are represented using one-hot encoding.

4.3.4 Implementation Environment

All experiments were conducted on a MacBook Pro equipped with an Apple M4 Pro processor and 48 GB RAM. Deep learning methods primarily utilized Apple MPS acceleration, with automatic fallback to CPU when instability was encountered.Statistical and machine learning methods were mainly implemented using scikit-learn, miceforest, and fancyimpute. Deep learning methods were implemented using PyTorch or adapted from the original authors' open-source implementations.All experimental results were stored incrementally in JSON format. In addition, a checkpoint–resume mechanism was implemented to allow interrupted experiments to continue from the last saved state.

4.4 Results Evaluation

To enable fair comparisons across datasets, this study ranks the 13 imputation methods based on imputation RMSE under each (dataset,mechanism,pattern,ratio,seed) condition, where 1 indicates the best performance and 13 indicates the worst. The average rank of each method across all conditions is then reported.A Friedman test is conducted to examine whether significant differences exist among the methods, revealing a statistically significant result ($x^2 = 8769.7, p < 0.001$). Subsequently, pairwise comparisons are performed using the Holm-corrected Wilcoxon signed-rank test. Among the 78 method pairs, 76 pairs show statistically significant differences at $\alpha = 0.05$.

Table 3: Hyperparameters for statistical and machine learning methods

Method	Key Hyperparameters
Mean	—
Median	—
EM	max_iterations = 100, convergence_threshold = 1e-4
MICE	max_iterations = 10, n_imputations = 5, estimator = BayesianRidge
KNN	n_neighbors = 5, weights = distance
MissForest	n_estimators = 100, max_iterations = 10
SoftImpute	max_rank = min(n_features, 50), max_iterations = 100
XGBoost	n_estimators = 100, max_depth = 6, learning_rate = 0.1, max_iterations = 10

Table 4:Hyperparameters for deep learning methods

Method	Key Hyperparameters
MIDA	hidden_layers = [256, 128, 256], epochs = 200, lr = 1e-3, corruption_rate = 0.5
MIWAE	latent_dim = 32, n_hidden = 128, n_samples = 20, epochs = 200, lr = 1e-3
GAIN	hidden = [256, 128], epochs = 300, lr = 1e-3, hint_rate = 100
TabCSDI	diffusion_steps = 50, hidden_dim = 128, n_layers = 4, epochs = 200, lr = 1e-3
MIRACLE	hidden_dim = 128, n_layers = 3, epochs = 200, lr = 1e-3, λ_reg = 1.0

Common DL settings: batch_size=256, optimizer=Adam, early_stopping_patience=10, device=MPS/CPU.

4.4.1 Overall Method Comparison

Table 5 presents the overall performance of the 13 imputation methods. MissForest achieves the best overall performance with an average rank of 2.06, obtaining the lowest RMSE in 9 out of the 11

Table 5: Overall method performance averaged across all experimental conditions.

Method	Family	Avg Rank	RMSE	MAE	WD	Time(s)
MissForest	Non-DL	**2.06**	**0.161**	**0.101**	**0.010**	79.18
XGBoost	Non-DL	2.71	0.170	0.108	0.013	71.60
KNN	Non-DL	4.49	0.196	0.129	0.014	10.40
EM	Non-DL	5.05	0.219	0.153	0.016	2.12
MICE	Non-DL	5.52	0.209	0.148	0.014	3.52
Mean	Non-DL	7.23	0.273	0.216	0.026	0.002
MIDA	DL	7.66	0.298	0.239	0.026	8.13
MIWAE	DL	7.69	0.298	0.232	0.028	17.42
SoftImpute	Non-DL	8.41	0.312	0.226	0.023	1.29
GAIN	DL	8.64	0.333	0.273	0.036	1.88
Median	Non-DL	9.22	0.304	0.226	0.027	0.003
MIRACLE	DL	10.66	0.381	0.307	0.034	21.00
TabCSDI	DL	11.66	0.849	0.681	0.085	9.50

All metrics: lower is better. WD = Wasserstein Distance. Avg Rank computed via within-condition ranking.

Table 6: Average imputation rank by missing mechanism.

Method	MCAR	MAR	MNAR	Δ(MNAR−MCAR)
MissForest	**2.04**	**2.50**	**1.65**	-0.39
XGBoost	2.59	2.70	2.84	+0.26
KNN	4.19	4.92	4.35	+0.16
EM	4.61	6.08	4.46	-0.15
MICE	5.84	5.93	4.80	-1.04
Mean	7.08	6.52	8.10	+1.02
MIDA	7.66	7.60	7.73	+0.07
MIWAE	7.43	7.12	8.52	+1.09
SoftImpute	8.82	8.74	7.66	**-1.16**
GAIN	8.96	8.29	8.66	-0.31
Median	9.13	8.57	9.97	+0.83
MIRACLE	10.68	10.40	10.88	+0.20
TabCSDI	11.96	11.62	11.38	-0.58

datasets. XGBoost ranks second with an average rank of 2.71, and both methods significantly outperform the remaining approaches (Wilcoxon $p < 0.001$).Overall, non-deep learning (non-DL) methods achieve a significantly better average rank (5.59) than deep learning (DL) methods (9.26). Notably, the imputation quality of several DL methods does not consistently surpass the Mean baseline. The average rank differences between MIDA, MIWAE, GAIN, and Mean are all within one rank, while MIRACLE and TabCSDI even rank lower than Median.The distribution preservation metric (Wasserstein Distance) shows a trend highly consistent with RMSE. MissForest (0.010) and XGBoost (0.013) achieve the lowest average WD values, whereas TabCSDI (0.085) exhibits the highest.

4.4.2 Effect of Missing Mechanism and Rate

Table 6 presents the average ranking of each method under the three missing mechanisms. Overall, RMSE values are substantially higher under the MNAR setting than under MCAR and MAR, confirming that non-random missingness poses the greatest challenge for imputation.However, the relative rankings among methods exhibit notable changes under MNAR. MissForest achieves an even better rank under MNAR (1.65) than under MCAR (2.04), suggesting that it possesses a certain degree of robustness to non-random missingness. In contrast, Mean Imputation drops from 7.08 under MCAR to 8.10 under MNAR, showing the largest performance degradation and reflecting the limitation of global statistics when missingness is not random. MICE also shows improved perfor-

Table 7: Imputation rank vs. downstream task rank.

Method	Family	Imp Rank	DS Rank	Gap
MissForest	Non-DL	**2.06**	**5.50**	+3.44
XGBoost	Non-DL	2.71	5.74	+3.03
KNN	Non-DL	4.49	6.81	+2.32
EM	Non-DL	5.05	7.08	+2.03
MICE	Non-DL	5.52	6.78	+1.26
Mean	Non-DL	7.23	6.23	-1.00
MIDA	DL	7.66	7.09	-0.57
MIWAE	DL	7.69	6.61	-1.08
SoftImpute	Non-DL	8.41	7.38	-1.03
GAIN	DL	8.64	6.43	-2.21
Median	Non-DL	9.22	7.15	-2.07
MIRACLE	DL	10.66	7.77	-2.89
TabCSDI	DL	11.66	8.35	**-3.31**

Gap = DS Rank − Imp Rank. Negative values indicate downstream performance better than imputation rank predicts.

mance under MNAR (5.84 → 4.80), possibly because its iterative conditional modeling can partially capture the dependency structure of missing values.

Regarding the impact of the missing rate, the relative rankings of the methods remain highly stable across the range of 10% to 50% missingness. MissForest and XGBoost consistently occupy the top two positions across all missing rates. Interestingly, the rank of Mean Imputation slightly improves at higher missing rates (7.71 → 6.64). This occurs because model-based approaches (e.g., EM and KNN) experience faster performance degradation as the available information decreases, thereby reducing their advantage over the Mean baseline.

4.4.3 Imputation Quality vs. Downstream Task Performance

Figure 3 reveals one of the most important practical findings of this study: a clear decoupling between imputation accuracy and downstream task performance. As shown in the figure, the imputation ranks and downstream ranks of the methods do not align along the diagonal but instead exhibit substantial deviations.

Specifically, as shown in Table 7, although MissForest ranks first in terms of imputation RMSE, it only ranks fifth in downstream task performance. In contrast, Mean Imputation achieves an imputation rank of sixth, yet its downstream performance (DS rank 6.23) surpasses that of several more sophisticated methods.This discrepancy arises because RMSE evaluates element-wise imputation accuracy, whereas downstream classifiers are more concerned with whether the structural relationships among features are preserved. Although Mean Imputation produces relatively coarse imputations, it does not introduce spurious feature correlations, and therefore tends to interfere less with downstream models.In contrast, several deep learning methods (e.g., GAIN and MIWAE) achieve substantially better downstream ranks than their imputation ranks. This suggests that the latent distribution learned by generative models may help preserve the decision boundaries required for downstream tasks, even when element-wise imputation accuracy is not optimal.

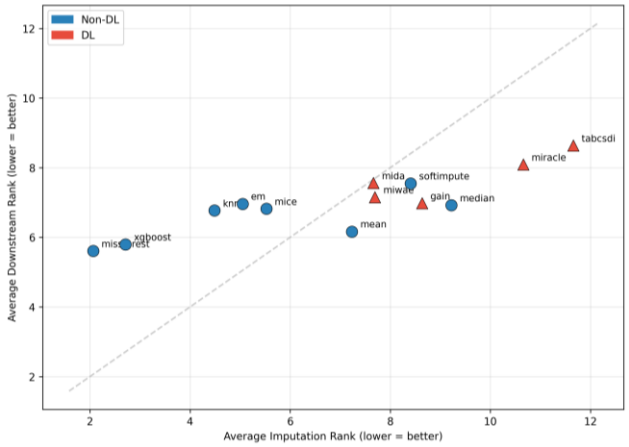


Figure 3: Imputation Quality vs. Downstream Task Performance

4.4.4 Computational Efficiency

In terms of computational efficiency, the runtime of different methods spans five orders of magnitude, ranging from Mean Imputation (0.002 s) to MissForest (79 s). Considering both imputation quality and computational cost, KNN (rank 4.49, 10.4 s) and MICE (rank 5.52, 3.5s) provide a favorable performance–efficiency trade-off.In contrast, deep learning methods generally fail to translate their additional computational cost into corresponding improvements in imputation quality.

4.5 Discussion

This section provides an in-depth discussion of the key findings derived from the experimental results and outlines the limitations of this study.

4.5.1 Dominance of Tree-Based Methods

MissForest and XGBoost consistently rank among the top two methods across nearly all experimental conditions. This finding

aligns with recent research trends in tabular data learning. In a comprehensive comparison presented at NeurIPS 2022, Grinsztajn et al.[20] reported that tree-based methods generally outperform deep learning approaches on small- to medium-sized tabular datasets. The main reason is that tree-based models possess a natural advantage in modeling nonlinear relationships and feature interactions, while also requiring less data to achieve strong performance.The results of this study extend this observation from supervised learning settings to the missing value imputation scenario, further supporting the versatility of tree-based methods in tabular data–related tasks.

4.5.2 Deep Learning Methods Underperform Expectations

The five deep learning (DL) methods achieve an overall average rank of 9.26, which not only lags significantly behind that of non-DL methods (5.59) but also indicates that most DL approaches fail to consistently outperform the Mean baseline. This outcome can be attributed to two main factors.First, dataset size limitations. The datasets used in this study range from 300 to 48,000 samples, with most containing fewer than 5,000 instances. Deep learning models typically involve substantially more parameters than traditional methods, making them more prone to overfitting or unable to adequately learn the underlying data distribution in small-sample settings.Second, the impact of fixed hyperparameters. To ensure fair comparisons, all methods in this study are evaluated using fixed hyperparameters without dataset-specific tuning. However, DL methods are generally more sensitive to hyperparameter choices—such as learning rate, hidden dimension, and training epochs—than tree-based approaches. Consequently, fixed configurations may underestimate the full potential of DL models. Nevertheless, the need for extensive hyperparameter tuning to achieve performance comparable to simpler methods itself reflects the deployment cost disadvantage of DL approaches in practical applications.

4.5.3 Analysis of TabCSDI Performance

TabCSDI ranks last among all 13 methods evaluated in this study (avg rank 11.66, RMSE 0.849). This result is not driven by a small number of difficult datasets. Across all 11 datasets, TabCSDI consistently ranks within the bottom three methods, and in 1,320 experimental runs, it ranks last in 51.7% of the cases.Compared with MissForest, the RMSE of TabCSDI can be 5 to 9 times higher on small datasets (e.g., 9.32× on Yeast and 8.82× on Spambase). Although this gap narrows on larger datasets—converging to approximately 1.3 to 2.5 times higher RMSE—TabCSDI still remains the worst-performing method overall. Further analysis reveals a significant negative correlation between TabCSDI performance and dataset size (Spearman $\rho = -0.74$, $p = 0.010$), indicating that its performance improves as the dataset size increases.This phenomenon can be explained by the structural challenges faced by diffusion models on small tabular datasets. TabCSDI is adapted from CSDI [18], which was originally designed for time-series data. In the original setting, the denoising process can leverage contextual information from adjacent time steps. However, when transferred to tabular data, this contextual advantage largely disappears.This analysis is consistent with existing literature. Kotelnikov et al.[21], in their study on TabDDPM, report that diffusion models struggle to outperform tree-based methods on small tabular datasets ($n < 5{,}000$), even when using more than 1,000 diffusion steps and carefully designed noise schedules. Moreover, the original CSDI [18]experiments were conducted primarily on time-series datasets with $n > 10{,}000$, and were not evaluated on small tabular datasets.

4.5.4 Decoupling Between Imputation Accuracy and Downstream Performance

This study reveals a clear decoupling between imputation accuracy (imputation RMSE rank) and downstream task performance (downstream F1 rank), as illustrated in **Figure 3**. Although MissForest ranks first in terms of imputation accuracy, it only ranks fifth in downstream task performance. In contrast, Mean Imputation ranks sixth in imputation accuracy but achieves a fourth-place ranking in downstream performance, outperforming several more complex methods.A possible explanation for this phenomenon is that RMSE measures element-wise imputation error, whereas downstream classifiers rely more heavily on whether the structural relationships among features and decision boundaries are preserved. Although Mean Imputation produces relatively coarse imputations, filling all missing values with the column mean does not introduce spurious feature correlations, and therefore causes less distortion to the decision boundaries of downstream models. In contrast, more sophisticated imputation methods may achieve higher element-wise accuracy but potentially alter the joint distribution structure among features.

Furthermore, several deep learning methods (e.g., GAIN and MIWAE) achieve noticeably better downstream ranks than their imputation ranks, suggesting that the latent distributions learned by generative models may help preserve information relevant to downstream tasks, even when element-wise imputation accuracy is not optimal.This finding carries important practical implications. Relying solely on imputation error as the criterion for selecting imputation methods is insufficient. Instead, the choice of evaluation metrics and imputation strategies should be aligned with the ultimate application objective, such as imputation accuracy, downstream predictive performance, or distribution preservation.

4.5.5 Limitations

This study has several limitations.First, the use of fixed hyperparameter settings may underestimate the optimal performance of certain methods, particularly deep learning approaches. Future work could explore the upper performance bounds of different methods through systematic hyperparameter sweeps.Second, the datasets used in this study range from 300 to 48,000 samples, with most falling into the small- to medium-sized category. Deep learning methods may exhibit different advantages on much larger datasets (e.g., millions of samples), which remains to be further investigated.Third, the MNAR simulation in this study generates self-dependent missingness probabilities using a logistic function. Although this approach follows standard practices in the literature, real-world non-random missingness often involves more complex dependency structures, such as multivariate interactions or external driving factors. Consequently, the external validity of the simulated results may be limited.Fourth, this study employs Random Forest as the sole downstream model. Different downstream models (e.g., linear models or neural networks) may have varying sensitivity to imputation errors. Therefore, relying on a single downstream model may not fully capture how imputation quality affects all types of downstream tasks.

5. CONCLUSION

This study presents a systematic literature review and empirical evaluation of missing value imputation methods, covering 13 representative approaches from traditional statistical methods, machine learning methods, and deep learning methods. Experiments were conducted on 11 public tabular datasets under three missing mechanisms (MCAR, MAR, MNAR), two missing patterns, and four missing ratios, resulting in a total of 17,160 experimental runs.

All methods were evaluated using default hyperparameter settings in order to better reflect realistic deployment scenarios.

The experimental results reveal three key findings. First, tree-based methods, particularly MissForest and XGBoost-based imputation, consistently achieve the best imputation quality across nearly all experimental settings, confirming the effectiveness of ensemble learning approaches for tabular data imputation tasks. Second, the five deep learning methods evaluated in this study do not consistently outperform simple baselines such as Mean Imputation under default hyperparameter configurations. This suggests that the advantages of deep learning approaches may only emerge with larger datasets or more carefully tuned hyperparameters. Third, and most importantly, this study identifies a clear decoupling between imputation accuracy and downstream task performance. Methods that achieve the lowest reconstruction error do not necessarily yield the best downstream predictive performance, while simpler approaches such as Mean Imputation can remain surprisingly competitive by preserving feature structures relevant to downstream decision boundaries.These findings carry important practical implications. When the primary objective is accurate reconstruction of missing values, tree-based imputation methods remain the most reliable choice under default settings. However, when the goal is to support downstream predictive tasks, practitioners should not rely solely on reconstruction metrics such as RMSE when selecting imputation methods. Instead, the evaluation criteria and imputation strategies should be aligned with the ultimate application objective, whether it is imputation accuracy, downstream predictive performance, or distribution preservation.

Future research could extend this work in several directions. First, systematic hyperparameter optimization could be conducted to better estimate the upper performance bounds of deep learning methods. Second, experiments could be expanded to larger-scale datasets to investigate whether deep generative models demonstrate stronger advantages in such settings. Third, more realistic MNAR simulation mechanisms could be explored to better reflect complex missingness processes in real-world data. Finally, future studies could examine how different downstream models respond to imputation errors, in order to better understand the interaction between imputation quality and downstream task performance.

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Table A1:Imputation RMSE by method and dataset (averaged across all missing settings and seeds).

Method	Adult	Autism	Bank	Breast	Concrete	Letter	Mammo.	Spam	Titanic	Wine	Yeast
MissForest	0.1581	0.1959	**0.1976**	**0.3199**	**0.1404**	**0.0963**	**0.0987**	**0.1175**	**0.1821**	**0.0903**	**0.1692**
XGBoost	**0.1537**	0.2013	0.2010	0.3625	0.1416	0.1107	0.1171	0.1244	0.1914	0.0911	0.1749
KNN	0.1774	0.2680	0.2319	0.3419	0.1683	0.1199	0.1281	0.1233	0.2944	0.1315	0.1742
EM	0.2004	0.3487	0.2430	0.3576	0.1781	0.1343	0.2028	0.1361	0.3285	0.0937	0.1821
MICE	0.1998	**0.1663**	0.2582	0.3623	0.1946	0.1449	0.2098	0.1268	0.3474	0.0989	0.1851
Mean	0.1879	0.3338	0.2402	0.5086	0.3232	0.2207	0.3115	0.1311	0.3692	0.1960	0.1833
MIDA	0.1842	0.5396	0.2314	0.4687	0.3062	0.2142	0.3145	0.1325	0.3600	0.2036	0.3234
MIWAE	0.1825	0.3338	0.2402	0.4701	0.3402	0.1266	0.2846	0.1290	0.5143	0.2390	0.4138
SoftImpute	0.3560	0.3338	0.2410	0.6230	0.3014	0.2369	0.3158	0.1387	0.4310	0.1409	0.3185
GAIN	0.1879	0.3524	0.2402	0.4876	0.3386	0.2502	0.3270	0.2748	0.3769	0.2821	0.5495
Median	0.1910	0.3451	0.2441	0.5986	0.3804	0.2164	0.4239	0.1381	0.4185	0.2008	0.1865
MIRACLE	0.3439	0.3271	0.3884	0.5734	0.4032	0.4104	0.3946	0.1309	0.4864	0.3291	0.4006
TabCSDI	0.3345	1.6172	0.2522	1.5584	1.0161	0.2405	0.3180	1.0371	1.1655	0.2239	1.5775

Bold indicates best per dataset.

Table A2: Imputation MAE by method and dataset (averaged across all missing settings and seeds).

Method	Adult	Autism	Bank	Breast	Concrete	Letter	Mammo.	Spam	Titanic	Wine	Yeast
MissForest	0.1089	0.1008	**0.1408**	**0.2390**	0.0802	**0.0659**	**0.0520**	0.0425	**0.1014**	**0.0628**	**0.1191**
XGBoost	**0.1077**	0.1028	0.1465	0.2701	**0.0798**	0.0806	0.0633	0.0498	0.1050	0.0652	0.1218
KNN	0.1278	0.1527	0.1667	0.2515	0.1054	0.0854	0.0704	**0.0417**	0.2032	0.0896	0.1244
EM	0.1460	0.2253	0.1858	0.2681	0.1357	0.1018	0.1206	0.0595	0.2442	0.0678	0.1228
MICE	0.1492	**0.0783**	0.1956	0.2821	0.1524	0.1116	0.1312	0.0596	0.2603	0.0728	0.1336
Mean	0.1425	0.2169	0.1881	0.4300	0.2875	0.1828	0.2603	0.0598	0.3149	0.1615	0.1337
MIDA	0.1398	0.4174	0.1798	0.3840	0.2613	0.1772	0.2617	0.0703	0.3030	0.1672	0.2686
MIWAE	0.1358	0.2169	0.1881	0.3763	0.2738	0.0973	0.2226	0.0586	0.4193	0.1969	0.3658
SoftImpute	0.2288	0.2169	0.1885	0.4758	0.2304	0.1864	0.2630	0.0647	0.3439	0.1109	0.1805
GAIN	0.1425	0.2410	0.1881	0.4199	0.2910	0.2107	0.2686	0.1591	0.3168	0.2412	0.5187
Median	0.1424	0.2247	0.1890	0.4654	0.2876	0.1758	0.3119	0.0562	0.3351	0.1649	0.1322
MIRACLE	0.2742	0.2113	0.3058	0.4655	0.3327	0.3729	0.3040	0.0602	0.4036	0.2781	0.3628
TabCSDI	0.2706	1.2819	0.1994	1.2492	0.8153	0.1980	0.2656	0.8247	0.9367	0.1822	1.2670

Bold indicates best per dataset.

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8. APPENDIX

The detailed results are provided in the appendix (**Table A1** and **Table A2**). Table A1 reports the imputation RMSE by method and dataset, while Table A2 presents the imputation MAE.

9. WORK DIVISION

Table 8: Work division Table.

Name	Work description	Contribution
Ren-He, Huang 114423025	1. Literature review on machine learning-based imputation methods 2. Experimental design and implementation 3. Drafting Sections 1, 2.3, 3, 4, and 5	40%
Chieh-Jung, Kuo 114423031	1. Literature review on deep learning-based imputation methods 2. Experimental design 3. Drafting Sections 1, 2.4, 3, 4, and 5 4. Document formatting and typesetting	40%
Chien-Chi, Chang 114423051	1. Literature review on traditional statistical imputation methods 2. Drafting Section 2.2	20%